



Tribhuvan University
Institute of Engineering
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Department of Electronics and Computer Engineering

GEO-LOCATING USERS IN MOUNTAINS BY DETECTING SKYLINES IN CAPTURED PHOTOS

Minor Project Proposal Presentation

Project Team

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Date June 16, 2026

Presentation Outlines

- 1 Introduction
- 2 Motivation
- 3 Objectives
- 4 Scope
- 5 Methodology
- 6 Expected Results
- 7 Applications

Why is this Project Needed?

- **GPS Failures in Mountains:** Standard GPS signals often fail in deep Himalayan valleys (like the Khumbu region) because steep peaks block the satellite signals.
- **Safety Hazards:** Getting lost in remote, high-altitude terrain can lead to life-threatening situations for trekkers and rescue teams.
- **Reliable Offline Backup:** A visual positioning system that runs offline on standard smartphones provides a critical navigation backup.
- **Natural Landmarks:** The mountain skyline is a permanent, distinct shape that can be used as a geometric reference for positioning.

Core Project Objectives

- **Generate Horizon Database:** Build a 360-degree database of virtual mountain views using digital elevation models (Copernicus DEM).
- **Develop Skyline Extractor:** Create a tool to detect and extract the mountain boundary line from photos, working even in haze or cloud cover.
- **Design Matching Engine:** Implement a fast, two-stage search that matches the photo's skyline against the database to estimate coordinates.
- **Validate with Real Data:** Test the system's accuracy using georeferenced Google Street View images along mountain trails.

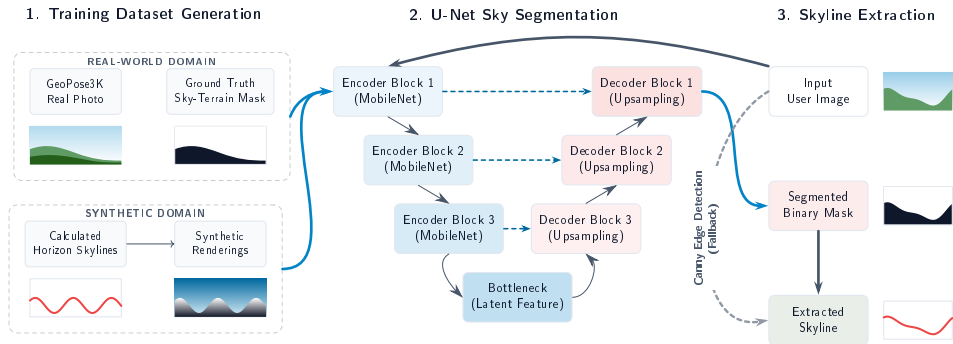
Scope and Limitations

- **Study Area (Khumbu Region):** Focuses on a $36.4 \text{ km} \times 31.2 \text{ km}$ area covering the main trekking routes in Nepal.
- **Coordinates of Interest:**
 - North-West: 86.600°E , 28.050°N
 - South-East: 86.970°E , 27.770°N
- **Offline Execution:** The system will run completely offline on standard Android mobile devices, requiring no cellular network.
- **Limitations:**
 - Relies on a visible skyline; fails in severe weather (dense fog) or if close obstacles (forests, walls) block the view.

Methodology Overview

- **Three-Part Pipeline:** The system consists of:
 - ① **Database Generation:** Ray-casting 3D elevation models to create virtual skyline shapes offline.
 - ② **Skyline Extraction:** Detecting and segmenting sky-mountain boundaries in user photos.
 - ③ **Matching Engine:** A fast two-stage search matching photo skylines to database profiles.
- **Key Approach:** Uses stable terrain features instead of fragile wireless networks or GPS signals.

Skyline Extraction Pipeline



Two-Stage Matching Engine



Working Principle

- **Ray-Casting Profiles:** Calculates the mountain peak angles for every direction (360°) from a grid of viewpoints.
- **Fourier-Accelerated Search (FFT):** Since the camera heading is unknown, we slide the photo skyline across the database. FFT completes millions of comparisons in under a second.
- **Elastic Shape Matching (DTW):** Lens distortions stretch photo shapes. Dynamic Time Warping adjusts the curves like an elastic band to align matching ridges.

System Instrumentation

Software Environment:

- **Development:** Python, PyTorch (deep learning), NumPy (array math).
- **Search Engine:** SciPy, FAISS (vector libraries).
- **Mobile Deployment:** Android NDK & OpenCV (offline on-device execution).

Hardware Environment:

- **Training Host:** Workstation with high-performance CUDA-enabled GPU.
- **Deployment Target:** Mid-range Android smartphone (offline search & validation).

Expected Deliverables

- **Offline Prototype App:** A mobile Android app that can calculate and show a user's location without any internet connection.
- **Map Visualization:** Displays the predicted coordinates on a local map along with the top 5 coordinate candidates.
- **Visual Skyline Overlay:** Overlays the photo's extracted skyline onto the 3D database view to let users verify the result.
- **Himalayan Database:** A pre-compiled 105 MB offline horizon database for the Khumbu region of Nepal.

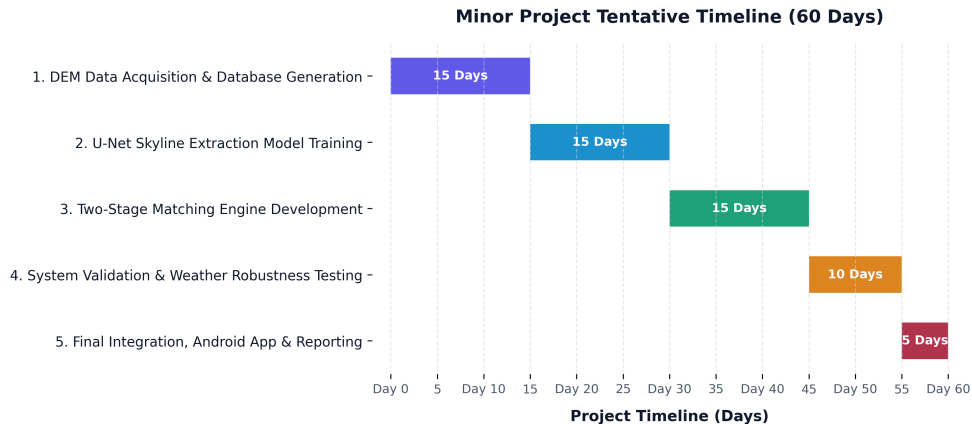
Target Accuracy & Validation

- **Testing Dataset:** System will be validated using 45,000 real panoramic images along the Everest Base Camp trail.
- **Location Accuracy:** Target successful grid cell matching within the 500-meter grid resolution.
- **Heading Accuracy:** Target camera direction estimation within a margin of $\pm 0.25^\circ$.
- **Weather Robustness:** Verification of matching performance under varying visibility levels (clear, hazy, and foggy profiles).

Project Applications

- **Search & Rescue (SAR):** Helps rescue teams locate lost trekkers or climbers in deep valleys where satellite/cellular networks fail.
- **Offline Tourist Navigation:** Serves as a reliable, battery-efficient offline guidebook/navigation tool for Himalayan tourists.
- **GNSS-Denied Navigation:** Acts as a robust positioning alternative in environments where GPS is blocked or denied.
- **Historical Imagery Alignment:** Can be used to orient historical outdoor photographs against terrain models to study glacier retreat or terrain changes.

Tentative Timeline



Estimated Project Budget

S.N.	Particulars / Activity	Cost (NRs.)
1	GPU Server Compute for Model Training	25,000
2	Data Storage and Street View Dataset	10,000
3	Map and Geocoding API Services	8,000
4	Android Testing Devices	15,000
5	Report Printing and Documentation	7,000
6	Contingency and Logistics	10,000
Total Estimated Budget		75,000

References I

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Thank You!

Questions & Answers

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